

Zain Khan

kn.zain@hotmail.com | linkedin.com/in/zain-khan | zainkhan.dev |  github.com/7kzaincode

EDUCATION

University of Waterloo

Bachelor of Applied Science (BASC), Computer Engineering

Waterloo, ON

Sep. 2025 – Apr. 2030 (Expected)

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C/C++, Java, C#, HTML/CSS, SQL

Frameworks & Libraries: React, Next.js (App Router), Node.js, Express.js, FastAPI, Tailwind CSS, Zustand, Chart.js, tldraw

Databases & Backend: PostgreSQL, MongoDB, Supabase, Firebase

Tools & Platforms: Git, Docker, Linux, Vercel, GitHub API, Vercel API, Edge Functions, Postman, Figma

Concepts: System Design, API Design, Data Structures & Algorithms, OOP, Concurrency, Async Pipelines, Performance Optimization, Automation, Auth (JWT/OAuth), Caching, Data Modeling

EXPERIENCE

WATonomous

Robotics Software Engineer — EVE Autonomy

Waterloo, ON

Jan. 2026 – Present

- Developed **concurrent autonomous robotics software in C++ (ROS 2 Humble)**, coordinating **5+ real-time nodes** across perception, planning, and control.
- Designed **modular ROS 2 node architecture** spanning perception, world modeling, planning, and control to enable isolated testing in simulation.
- Implemented **costmaps, A* path planning, and Pure Pursuit control**, achieving stable point-to-point navigation in cluttered simulated environments.
- Debugged and visualized live robot state using **Foxglove** within **Docker**-based development environments.

Kasperia

Software Developer (Frontend & Data) — Contract

Remote

Jun. 2023 – Sep. 2025

- Improved page load performance by **30%** and reduced maintenance overhead by **25%** through reusable React components and modular CSS architecture.
- Built **automated data pipelines** using JavaScript, SQL, and REST APIs to aggregate KPI and traffic data, reducing manual reporting time by **40%**.
- Developed KPI dashboards with React and Chart.js, increasing stakeholder engagement by **35%** and supporting data-driven decisions.

PROJECTS

Sketchy | Next.js, TypeScript, Gemini 2.5 Flash, tldraw, MongoDB, GitHub OAuth, Vercel API

- Built a sketch-to-site web app with a **tldraw** canvas that converts hand-drawn wireframes into deployable HTML/CSS/JS in **under 60 seconds** end-to-end.
- Optimized a **Gemini 2.5 Flash inference pipeline** using constraint-driven prompting and schema validation; reduced tokens **3000 → 800 (73%)** while improving determinism.
- Implemented an **end-to-end CI-style deployment pipeline** integrating GitHub and Vercel APIs to generate repos, push source, and return live URLs.
- Added **GitHub OAuth** authentication and a **MongoDB Atlas** persistence layer supporting autosave, versioning, and rollback.

inven[s]tory | React, Next.js, TypeScript, Supabase, PostgreSQL, Tailwind

- Developed a **personal inventory and P2P exchange platform** with authenticated users and item lifecycle workflows.
- Designed **normalized PostgreSQL schemas** with Supabase-backed authentication and access controls to unify users, items, and exchanges.

Arcade Cabinet System | Raspberry Pi, Python, JavaScript

- Built a Raspberry Pi arcade system with optimized GPIO input handling, achieving **20% faster input response** under load.
- Repurposed the system into a **self-sustaining fundraising platform** for a school coding club.